

Bhoomika R

Embedded Software Engineer



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Bengaluru, Karnataka, India

Professional Summary

AUTOSAR Embedded Engineer with 3+ years of experience in automotive ECU development, specializing in Classic AUTOSAR (BSW, Diagnostics, SecOC, Communication Stack, CDD). Strong expertise in CAN/UDS debugging, ECU integration, and end-to-end V-model development. Proven track record in improving test efficiency, ensuring ASPICE compliance, and delivering high-quality automotive software for OEM projects.

Core Competencies

- AUTOSAR Classic (BSW, RTE, CDD, SecOC)
- Diagnostics (UDS – ISO 14229, DEM, DCM, DID, RID)
- CAN Communication & Debugging (Vector CANoe)
- Embedded C Development
- ECU Integration & Validation
- Test Automation (Python)
- ASPICE (SWE1 - SWE6) Process Compliance
- Static Code Analysis (MISRA-C, CERT-C, HIS)
- Requirement Engineering (DOORS, JIRA)

Technical Skills

- **Programming Languages:** C, Embedded C, Python, UML Scripting
- **AUTOSAR:** Classic AUTOSAR, BSW, Diagnostic Stack, Communication Stack, SecOC, CDD
- **Communication Protocols:** CAN, SPI, I2C, UART, LIN, UDS (ISO 14229)
- **Tools:** Vector CANoe, EB Tresos, WinIdea, SWATT, PLAST
- **Debugging:** CAN Trace Analysis, ECU Debugging, Test Bench Setup
- **Microcontrollers:** ARM Cortex, ST Chorus 10M
- **Development :** Git, GitHub, Jenkins, Visual Studio
- **Modeling & Documentation:** Enterprise Architect, Doxygen
- **Testing:** Unit Testing, Integration Testing, SIL/MIL, Regression Testing
- **Standards:** ASPICE, SDLC, Agile
- **Additional Exposure:** Basic knowledge of ISO 26262, RTOS concepts

Professional Experience

Continental Automotive Components (Aumovio) - Bengaluru, India

Jun 2022 – Present

Project: In Car Application Server (ICAS) - Gateway ECU (Volkswagen)

Key Responsibilities:

- Worked as an Embedded Software Developer in a Classic AUTOSAR platform software team, contributing to end-to-end software development and testing of modules.
- Developed and integrated AUTOSAR BSW modules including Diagnostics (DCM, DEM), SecOC and CDD components for gateway ECU.
- Configured and implemented UDS diagnostics services (DID, RID) improving ECU diagnostics and fault handling.
- Performed advanced CAN debugging using Vector CANoe, analysing CAN traces and UDS responses to resolve communication issues.
- Worked on Secure Onboard Communication (SecOC) ensuring secure ECU communication.
- Configured AUTOSAR models using EB Tresos and validated ECU behaviour on target hardware and software.
- Built and maintained ECU Test Bench environments for software validation and debugging activities.
- Developed Software Requirements, developed and executed Unit Tests, Software Integration Tests and Software Qualification Tests.
- Achieved high MC/DC coverage through unit testing using SWATT, improving software reliability.
- Developed Python scripts for test automation, improving validation efficiency and reduced manual testing effort.
- Contributed to ASPICE Level 2 compliance, supporting SWE.1 to SWE.6 lifecycle activities, including requirement traceability, design reviews and test coverage.
- Collaborated closely with cross-functional teams (system, testing, quality) for defect resolution and successful software delivery.

Key Achievements:

- Contributed to successful ASPICE Level 2 assessment.
- Improved testing efficiency through automation scripts, reducing manual effort.

Education

K S School of Engineering and Management - Bengaluru, India

2017–2021

Bachelor of Engineering in Electronics and Communication

CGPA: 7.2 / 10

Certifications

- Advanced Embedded System course - Vector India Pvt Ltd
- AUTOSAR course - Vector India Pvt Ltd